

- Q.31 Draw a neat sketch of four bar chain.
- Q.32 Differentiate between lower pair and higher pair with an examples.
- Q.33 A shaft carries four masses A, B, C and D of 12 kg, 10kg, 18kg and 15kg respectively. The mass rotate in the same plane and their radius of rotation are 40mm, 50mm, 60mm and 30mm respectively. The angular position of masses B, C and D are 60 deg, 135 deg and 270deg from the mass A. determine the magnitude and position of the balancing mass at a radius of 100mm.
- Q.34 Name the different type of gear train. Explain any two with diagram.
- Q.35 Draw the turning moment diagram of four stroke single cylinder IC engine.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain different types of constrained motions with an example.
- Q.37 Explain gear nomenclature with the help of a neat sketch.
- Q.38 Explain the inversion of single slider crank chain with the help of neat sketches.

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6th Sem / Mechatronics Sub : Mechanisms & Machines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Spring is a type of
a) Fluid link b) Rigid link
c) Flexible link d) None of the above
- Q.2 Quick return motion mechanism is an inversion of
a) Slider crank chain
b) Double slider crank chain
c) Double lever mechanism
d) None of the above
- Q.3 Which of the following gear system have minimum axial thrust?
a) Spur b) Helical
c) Bevel d) Double helical
- Q.4 A belt drive is
a) A positive drive b) Not a positive drive
c) Parallel axis d) None of the above
- Q.5 Crowning of pulley is done to
a) Prevent the belt running off the pulley
b) Improve the shape of pulley form safety considerations
c) Improve the strength of pulley
d) Improve the torque transmitted by the pulley

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- Q.6 Unbalance in rotating parts may be due to
- Un-machined portion of castings
 - Lack of homogeneity in the material
 - Non symmetry of parts
 - Any of the above
- Q.7 Longitudinal vibration are those in which particles of body move
- Parallel to the axis
 - About the axis
 - Perpendicular to the axis
 - None of the above
- Q.8 The ratio of maximum fluctuation of energy to work done per cycle is called
- Co-efficient of fluctuation of speed
 - Co-efficient of fluctuation of energy
 - Fluctuation of speed
 - None of the above
- Q.9 Practical application of cam are in
- Gear cutting machine
 - Printing machine
 - Sewing machines
 - All of the above
- Q.10 The pulley and belt in belt drive act as
- Cylindrical pair
 - Rolling pair
 - Turning pair
 - None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Belt, ropes, chains etc. may be considered as a resistant bodies. State (True / False)
- Q.12 Neutral section is consideration at _____ the thickness of the belt.

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- Q.13 Define static balancing.
- Q.14 Cam and follower is an example of closed pair. State (True / False)
- Q.15 To and fro motion of a particle about a fixed point is called _____.
- Q.16 Slip is generally expressed in _____.
- Q.17 Flywheel is used in electric motor, turbine etc. State (True / False)
- Q.18 The SI unit of mass moment of inertia is _____.
- Q.19 In case of knife edge follower, the base circle and prime circle are _____.
- Q.20 Cams are generally manufactured by a punch, die casting or milling (True / False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Enlist the various advantages of V-belts over flat belts.
- Q.22 Differentiate between machine and structure.
- Q.23 Derive an expression for fluctuation of energy in flywheel.
- Q.24 Write a short note on cam and follower. Also list its various functions.
- Q.25 Define link. Explain its different types.
- Q.26 Explain crowning of pulley in details.
- Q.27 Describe in details the static balancing with example.
- Q.28 What is a gear? Explain its various type.
- Q.29 Define the following with reference to cam
- angle of ascent
 - Dwell
 - prime circle.
- Q.30 Why is rim type of flywheel preferred over disc type? Give reasons.

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